

AI-POWERED CAREER & SKILL GAP ANALYZER

Unit 1 Mini Project - Product Engineering & Design Thinking

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Project Vision

Help students understand where they are, where they want to go, which skills are missing, and what they should do next through a transparent, personalized and measurable career-planning workflow.

This document is a fresh project report created specifically for the topic AI-Powered Career & Skill Gap Analyzer. The reference material is not inserted into this PDF.

1. Contents and Project Map

Section	Coverage
Project overview	Problem, objectives, scope and workflow
Product engineering	SDLC, design principles and project boundaries
Design Thinking	Empathize, Define, Ideate, Prototype and Test
User research	Interview plan, empathy map and journey
Personas	Student, mentor, admin and engineering stakeholders
Prototype	Dashboard, assessment, gap analysis and roadmap
Requirements	Functional and non-functional requirements
MVP	Vision, features, risks and release plan
Architecture	Application layers, database and security
Agile + DevOps	Sprints, repository, CI/CD, Docker and cloud roadmap
Verification	Test cases, acceptance criteria and demonstration flow

Core workflow

Student profile -> target career -> skill assessment -> skill-gap analysis -> personalized roadmap -> progress tracking -> reassessment

2. Abstract, Objectives and Scope

Abstract. AI-Powered Career & Skill Gap Analyzer is a proposed web platform that helps students connect their current skills and evidence to a target career role. The platform combines structured career data, skill assessments and a recommendation workflow to identify gaps and generate an actionable learning roadmap.

The project is designed as a product engineering exercise, not only as an AI model. It therefore covers user research, Design Thinking, prototype design, requirements, architecture, data modelling, testing and a DevOps roadmap.

Objectives

- Create a structured student profile with education, skills, projects and career interests.
- Provide a career-role catalogue with required skills and proficiency expectations.
- Assess current skill levels using questions and evidence.
- Compare current skills with target-role requirements and identify gaps.
- Generate a prioritized learning roadmap.
- Track progress and support reassessment over time.
- Build a secure and maintainable architecture that can later be deployed to the cloud.

Scope boundaries

- MVP focuses on one student workflow and a curated career/skill catalogue.
- Resume parsing, external job feeds, advanced labor-market forecasting and AI interview coaching are future expansions.
- The platform provides decision support; it does not guarantee employment or predict a person's future career outcome.

3. Problem Statement and Business Justification

Students can accumulate certificates, courses, projects and programming skills without a single place that explains how those capabilities relate to a chosen career role. Career information is often scattered across job descriptions, course sites, social media and informal advice.

Problem statement

Students need a transparent way to compare their current skills with the requirements of a target career because generic learning lists do not show which missing skills matter most or what sequence of actions is practical for their starting point.

Business justification

Activity	Current difficulty	Proposed support
Career selection	Role descriptions can feel broad	Structured career-role profiles
Skill inventory	Skills and evidence are scattered	Central skill profile
Assessment	Self-rating can be inconsistent	Assessment + evidence
Gap analysis	Hard to compare current vs target	Priority gap view
Learning planning	Too many resources, unclear order	Personalized roadmap
Progress	Learning activity is not connected to goals	Milestones and progress tracking
Reporting	Manual summaries	Dashboard and readiness report

4. Product Engineering and SDLC

Product engineering considers the full journey from understanding a user problem to building, testing, deploying and improving a reliable product.

Aspect	Programming	Product engineering
Goal	Implement a function	Solve a user problem reliably
Work	Code and debugging	Research, design, code, testing and operations
Success	Correct execution	Usability, correctness and maintainability

Proposed SDLC

1. User research -> 2. Requirements -> 3. Design Thinking -> 4. Prototype -> 5. Architecture and database -> 6. Implementation -> 7. Testing and CI/CD -> 8. Deployment and monitoring -> 9. Feedback and iteration

Project risks caused by unclear requirements or premature technology choices are reduced by keeping the user problem and MVP explicit before implementation.

5. Design Thinking - Overview

Design Thinking - Five Stages



CareerPath AI application

Student interviews -> skill-gap problem -> solution ideas -> dashboard/assessment/roadmap prototype -> usability testing

The cycle repeats whenever testing reveals a better problem definition or design.

CareerPath AI uses five stages

Stage	Purpose	Artifact
Empathize	Understand real user needs and pain points	Research plan, interview questions, empathy map
Define	Turn observations into one focused problem	Problem statement, HMW question
Ideate	Generate and prioritize possible solutions	Idea matrix, selected concept
Prototype	Make the proposed experience visible	UI flows and prototype screens
Test	Observe how people use the design	Task test, feedback log, iteration plan

The stages are iterative. Test findings can change the problem definition or solution idea.

6. Stage 1 - EMPATHIZE

Empathize means learning how students actually make career decisions before choosing a solution or model.

Research plan

- Interview 3-5 students with different experience levels.
- Interview 1 faculty member, career coordinator or mentor where available.
- Observe how a student currently chooses a target role and plans learning.
- Do not collect passwords or unnecessary personal information.
- Record only evidence relevant to the problem: task, friction, expectation and desired improvement.

Interview questions

Question	What it helps us learn
How do you choose a career role?	Current decision process
How do you know what skills you already have?	Skill evidence and confidence
How do you decide what to learn next?	Planning behavior
What is confusing about job descriptions?	Information gaps
What makes a recommendation trustworthy?	Trust and transparency
What information would you prefer not to share?	Privacy expectations

Research status

The interview examples in this report are a proposed research plan. They should be replaced with real observations, quotes and findings after field validation.

7. Stage 2 - DEFINE

The Define stage compresses research into a specific user problem that the MVP can address.

How Might We

How might we help students understand their readiness for a target technical career and identify a practical sequence of skills to develop next?

Point of view

A motivated student needs an understandable link between current evidence and target-role expectations because generic lists of skills do not explain which missing abilities should be developed first.

Design principles

- Explain why a skill was identified as a gap.
- Make the first user journey short and clear.
- Allow students to correct or update self-reported information.
- Separate self-reported skill from supporting evidence.
- Treat the roadmap as guidance rather than a guarantee.
- Protect student information with role-based access and minimal data collection.

8. Stage 3 - IDEATE

The Ideate stage explores multiple solution concepts before selecting the MVP.

Idea	User value	Complexity	Release
Skill-gap analyzer	High	Medium	MVP
Personalized roadmap	High	Medium	MVP
Career explorer	High	Low	MVP
Resume analyzer	Medium	Medium	Later
GitHub project analyzer	Medium	High	Later
AI interview coach	Medium	High	Future
Job recommendation	Medium	High	Future
Mentor review	Medium	Medium	Later

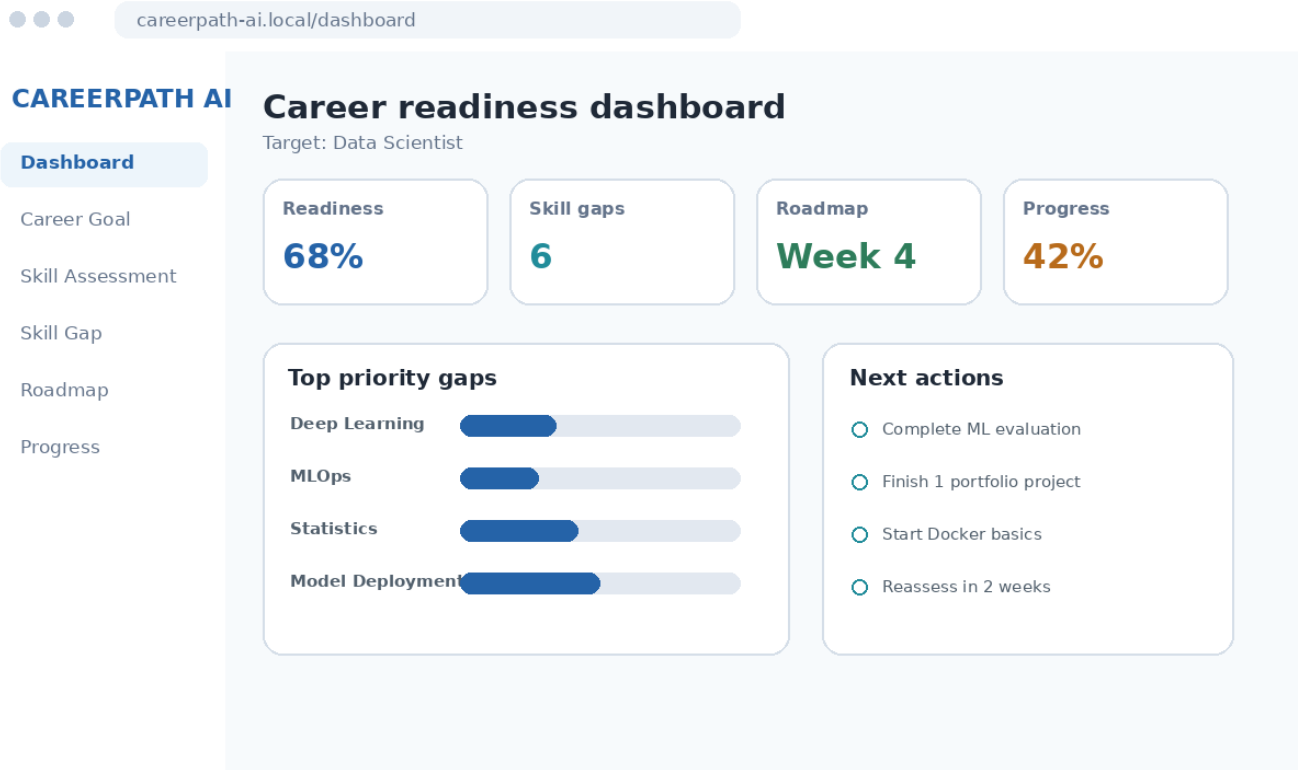
Selected solution

A browser-based Career and Skill Gap platform with student, mentor and admin views, backed by a shared database and an explainable gap-analysis engine.

Why this solution fits the MVP

- It directly addresses the defined user problem.
- The core logic can be verified with deterministic rules before advanced ML is introduced.
- The workflow creates measurable outputs: gap list, roadmap and progress state.
- The architecture can evolve into a larger AI-enabled platform without changing the core domain model.

9. Stage 4 - PROTOTYPE: Student Dashboard



Prototype goals

- Show the student's target career first.
- Provide a quick readiness summary without overwhelming the user.
- Make the top priority gaps visible.
- Provide clear next actions.
- Give direct navigation to assessment, gaps and roadmap.

Prototype status: illustrative static design, not a screenshot of a running application.

10. Prototype: Skill Assessment and Evidence

Skill Assessment

Rate your current ability and attach evidence where possible.

Python

Self-rating: Strong



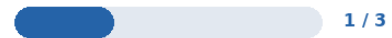
SQL

Self-rating: Good



Statistics

Self-rating: Developing



Machine Learning

Self-rating: Good



Deep Learning

Self-rating: Beginner



Evidence example: project link, internship task, certificate, code sample, or assessment result.

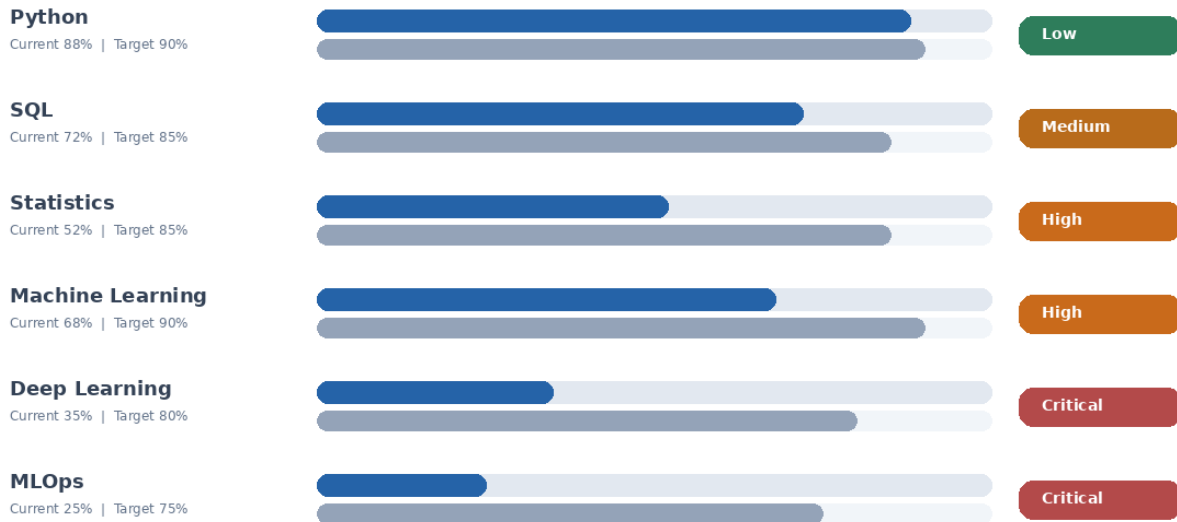
Assessment design

- Each skill should have a documented definition and expected proficiency levels.
- The student can provide evidence such as a project, certificate, course, internship task or coding sample.
- The system should distinguish confidence from evidence strength.
- The initial version can combine assessment score and evidence using documented, deterministic rules.
- Future versions may adapt question difficulty based on previous answers.

A transparent scoring explanation is more useful than an unexplained AI-generated score.

11. Prototype: AI Skill Gap Analyzer

Skill Gap Analysis - Data Scientist



Interpretation: prioritize high-gap, high-importance skills first.

Gap interpretation

Priority	Meaning	Example action
Critical	Large gap in an important target skill	Place early in roadmap
High	Significant development need	Schedule focused learning
Medium	Useful improvement	Add after critical gaps
Low	Close to target level	Maintain and practice

The percentages displayed above are illustrative UI data used to demonstrate the design, not the measured profile of a real person.

12. Prototype: Personalized Career Roadmap

Personalized 12-Week Career Roadmap

01	Statistics Probability, distributions, hypothesis testing	Next >
02	Machine Learning Evaluation, tuning, pipelines	Next >
03	Deep Learning Neural networks, regularization	Next >
04	Portfolio Project End-to-end ML project with API	Next >
05	MLOps Docker, CI/CD, deployment basics	Next >
06	Career Readiness Resume, GitHub, mock interview	Next >

Roadmap behavior

- Prioritize the highest-value gaps first.
- Break large skills into smaller milestones.
- Attach practice and portfolio evidence to milestones.
- Let the student update completion state.
- Reassess after a defined learning period and update the gap view.

Example roadmap logic: identify target skills -> calculate gaps -> rank by importance and gap -> create milestones -> attach resources/projects -> track progress -> reassess.

13. Stage 5 - TEST

Testing asks whether users can understand and complete the intended tasks without being coached through the interface.

Task	Expected action	Success target
Set target career	Find Career Goal and choose a role	Complete without assistance
Complete assessment	Answer all required items	Result is saved
Find a priority gap	Open Skill Gap and identify top item	Correctly identifies priority
Understand gap	Open details	Can explain why the gap matters
Find next action	Open Roadmap	Can identify next milestone
Update progress	Mark a milestone completed	Status changes and persists

Test evidence should record participant role, date, task, observed outcome, time, confusion and resulting design change.

14. Prototype Feedback and Iteration

Observed issue	Likely cause	Design response
User cannot find Skill Gap	Navigation or naming	Promote Skill Gap in dashboard and primary navigation
User questions the score	Insufficient transparency	Add scoring explanation and evidence details
User does not know what to learn first	Priority unclear	Add explicit priority and next-action labels
Roadmap feels too long	Information overload	Show current milestone first and allow expansion
User worries about personal data	Privacy unclear	Add concise data-use explanation and controls

Observe -> Reflect -> Make

Observe the user's behavior, reflect on why the friction occurs, make a focused change, then test the same task again. This makes Design Thinking an engineering loop rather than a one-time presentation topic.

15. Personas

Persona	Goal	Pain point	Design response
Aarav - Student	Prepare for Data Scientist role	Unclear priorities	Gap analysis + roadmap
Meera - Mentor	Guide multiple students	Different needs	Progress + gap overview
Dr. Patil - Admin	Maintain career and skill catalogue	Manual inconsistency	Role/skill management
Developer / QA	Build and verify product	Changing requirements	Documented requirements + test plan

Personas are design hypotheses until validated through real user research.

16. Stakeholders and Responsibilities

Stakeholder	Responsibility	Contribution
Students	Primary users	Validate problems, flows and usefulness
Mentors / Faculty	Domain reviewers	Validate career and skill definitions
Admin	Catalogue and access owner	Manage roles, skills and policies
Developer	Product implementation	Build frontend, backend and logic
QA	Quality verification	Execute functional and access tests
DevOps	Delivery and reliability	Automate builds, deployments and monitoring

17. Functional Requirements

ID	Requirement	Acceptance condition
FR01	Authentication and access	Registered users can log in; server enforces role permissions.
FR02	Student profile	Student can create and update education, interests and goal.
FR03	Career catalogue	Authorized users can view and manage career roles and required skills.
FR04	Skill catalogue	Skills have definitions, categories and target levels.
FR05	Assessment	Student can complete assessment and receive a stored result.
FR06	Skill-gap analysis	System compares current and target levels using documented rules.
FR07	Roadmap	System generates a prioritized roadmap linked to identified gaps.
FR08	Progress tracking	Student can update milestone status and view progress.
FR09	Reports	Authorized users can view basic readiness and progress summaries.
FR10	Auditability	Important administrative changes are recorded for review.

18. Access Control and Non-Functional Requirements

Action	Student	Mentor	Admin
View own profile	Yes	Yes	Yes
Complete assessment	Yes	Optional	No
View own gap and roadmap	Yes	Yes for assigned students	Yes
Manage career roles	No	Review	Yes
Manage skill catalogue	No	Review	Yes
View system reports	Limited	Assigned scope	Yes

Quality	Proposed requirement	Verification
Security	Password hashing, server authorization, HTTPS in deployment	Credential and access tests
Performance	Define measurable API target	Repeatable API/load test
Availability	Health checks and recovery process	Availability measurement
Maintainability	Separate modules and documented config	Architecture/code review
Data reliability	Validation, backup and recoverable storage	Consistency and restore test
Usability	Readable screens and clear labels	Observed user tasks
Privacy	Least-privilege access and minimal data collection	Data-flow review

19. Product Vision, MVP and Release Plan

Vision

Provide students with a personalized and understandable path from current skill evidence to a target career, while keeping recommendations explainable and progress measurable.

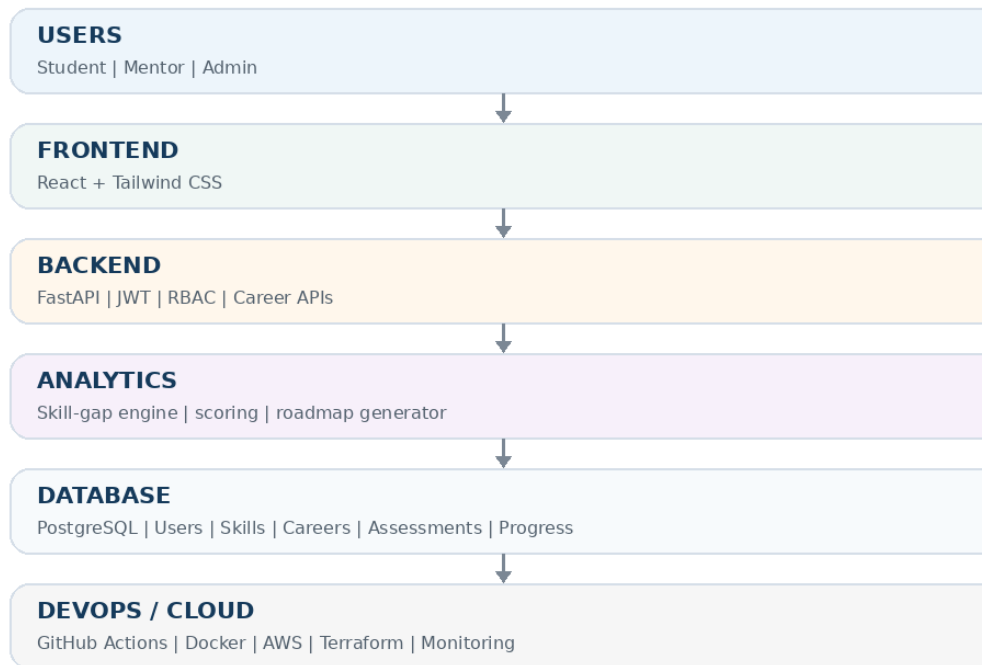
Capability	MVP	Later expansion
Profile	Education, interests, target role	Advanced portfolio profile
Career roles	Curated role catalogue	External trend updates
Skills	Definitions and target levels	External taxonomy integrations
Assessment	Questionnaire + evidence	Adaptive assessment
Gap analysis	Rule-based priority gaps	Evidence-weighted ML scoring
Roadmap	12-week structured plan	Dynamic adaptive roadmap
Progress	Milestones and completion	Learning-provider integration

Risks and mitigations

Risk	Impact	Mitigation
Scope growth	Incomplete MVP	Freeze core workflow
Weak skill definitions	Poor gap results	Review definitions with domain mentors
False precision	Over-trust in score	Explain score and show evidence
Privacy leak	Sensitive data exposure	Least privilege + secure storage
Model complexity too early	Hard to debug	Start with deterministic rules

20. System Architecture

AI-Powered Career & Skill Gap Analyzer - Architecture



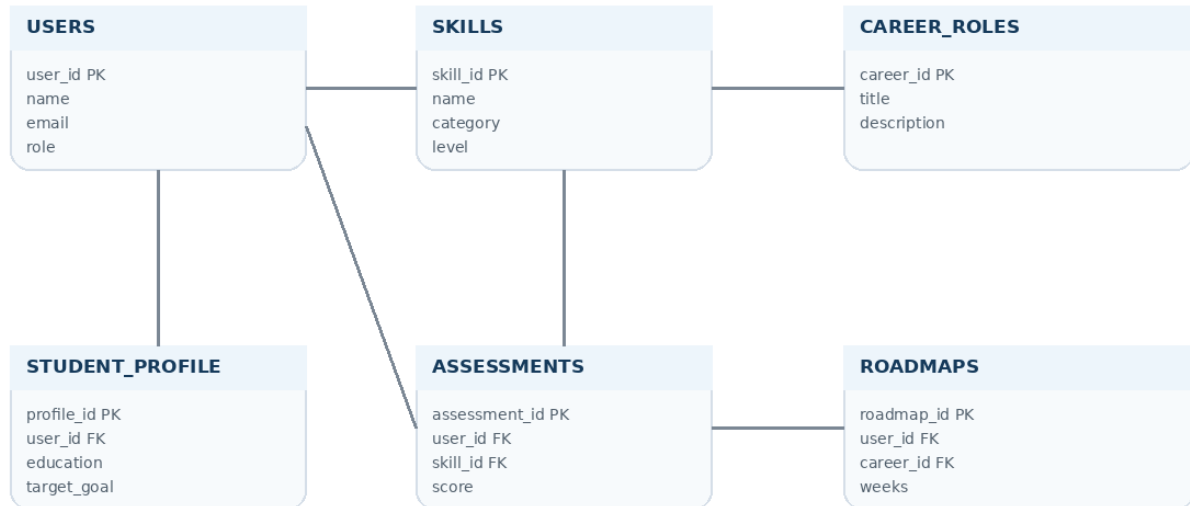
The browser never accesses the database directly; backend services enforce identity, permissions and business rules.

Technology stack

Layer	Technology	Purpose
Frontend	React + Tailwind CSS	Responsive web experience
Backend	Python FastAPI	REST APIs and business logic
AI / Analytics	Python, Pandas, Scikit-learn	Scoring and future predictive models
Database	PostgreSQL	Users, skills, careers, assessments, roadmaps
Authentication	JWT + secure password hashing	Identity and role-based access
DevOps	GitHub Actions + Docker	Build and test automation
Cloud	AWS + Terraform (future)	Infrastructure and deployment
Monitoring	Prometheus + Grafana (future)	Operational metrics

21. Database and Data Design

Core Data Model



PK = primary key; FK = foreign key. The logical model can be refined during implementation.

Entity	Purpose
USERS	Identity, email and role
STUDENT_PROFILE	Education, interests and target goal
SKILLS	Reusable skill definitions and categories
CAREER_ROLES	Target roles and required skills
ASSESSMENTS	Assessment sessions and skill results
ROADMAPS	Personalized learning plans
PROGRESS	Milestone completion and evidence
AUDIT_LOGS	Important administrative activity

The diagram is a logical design for Unit 1 and should be refined during database implementation.

22. Skill-Gap Analysis Logic

The first implementation can use a transparent rule-based engine. A more advanced ML model can be added after the product has reliable user and evidence data.

Proposed logic

Step	Input	Output
1. Identify target role	Selected career	Target skill set
2. Collect current state	Assessment + evidence	Current skill scores
3. Compare levels	Current vs target	Gap per skill
4. Apply importance	Skill importance weight	Priority score
5. Rank gaps	Priority + gap magnitude	Ordered gap list
6. Build roadmap	Gap list + time horizon	Learning milestones
7. Track progress	Completed milestones	Updated progress
8. Reassess	New evidence	Updated readiness view

Example conceptual formula

Gap = $\max(0, \text{Target Level} - \text{Current Level})$

Priority Score = Gap x Skill Importance

Higher priority scores are placed earlier in the roadmap.

These formulas are project rules for the MVP and should be validated against real user feedback before being treated as a production recommendation model.

23. Agile Plan and Engineering Roadmap

Sprint	Main work	Review output
1	Requirements, schema, authentication and roles	Role-controlled login and approved schema
2	Profile, career roles, skills and assessment	Working profile and assessment flow
3	Gap engine, dashboard and roadmap	End-to-end readiness demonstration
4	Progress, tests, documentation and DevOps starter	Reviewed local MVP and automated checks

Engineering roadmap

Requirements -> UI prototype -> Architecture -> Implementation -> Automated tests -> CI/CD -> Docker -> Infrastructure as code -> Cloud deployment -> Monitoring -> Continuous improvement

Only the stages actually completed should later be marked as implemented or deployed.

24. GitHub Repository and Execution Guide

Path	Purpose
frontend/	React user interface
backend/	FastAPI APIs and business logic
ml-engine/	Skill-gap and analytics logic
database/	Schema, migrations and seed data
tests/	Unit, integration and API tests
docs/	Requirements, diagrams, prototype and decisions
.github/workflows/	CI workflows
docker/	Container definitions
terraform/	Cloud infrastructure
monitoring/	Metrics and dashboards
README.md	Objective, scope, setup and architecture

Starter commands

```
mkdir career-skill-gap-analyzer
cd career-skill-gap-analyzer
git init
git add .
git commit -m "Initialize Unit 1 project"
```

25. Verification and Acceptance Plan

Test	Scenario	Expected result
T01	Correct and incorrect login	Correct credentials succeed; incorrect credentials rejected
T02	Student accesses another student's data	Access denied by backend
T03	Complete assessment	Result stored and shown
T04	Skill-gap calculation	Output matches documented rule
T05	Roadmap generation	Roadmap contains priority gaps and milestones
T06	Update progress	Milestone state changes and persists
T07	Unauthorized admin action	Server rejects request
T08	Backup restore	Core records recover
T09	Invalid input	Useful validation message
T10	Performance test	Measured result recorded against target
T11	Prototype task test	User can complete task without coaching

26. Suggested Demonstration Flow

A clear classroom demonstration should follow the real user journey rather than jumping between unrelated screens.

Step	Demonstration
1	Introduce the problem: students need clarity about career readiness and skill gaps.
2	Create or open a student profile.
3	Choose a target role such as Data Scientist.
4	Complete a sample skill assessment.
5	Open the Skill Gap view and explain one priority gap.
6	Open the personalized roadmap and show the next milestone.
7	Mark a milestone complete and show progress update.
8	Explain the architecture, database and DevOps roadmap.
9	Show tests and explain what is implemented versus planned.

Do not present a static prototype as a running system. Demonstration evidence should match what has actually been implemented.

27. Conclusion and Next Phase

AI-Powered Career & Skill Gap Analyzer is designed as a practical AI-DS project that connects product engineering, Design Thinking, software architecture and DevOps foundations.

The Unit 1 design establishes a coherent path from user research to a testable MVP. The most important next activity is validation: collect real student feedback, refine the career and skill definitions, then implement the smallest complete workflow.

Recommended implementation order

- Build authentication and student profile.
- Add career and skill catalogue.
- Implement assessment and rule-based gap engine.
- Create roadmap and progress workflow.
- Add automated tests and GitHub Actions.
- Containerize with Docker and prepare cloud deployment.
- Add monitoring and iterate based on user feedback.

UNIT 1 STATUS

Area	Status
Problem definition	Prepared
Design Thinking stages	Prepared; field validation pending
Prototype	Prepared as static illustrative designs
Requirements	Prepared
Architecture and database	Proposed
Testing	Planned; execution pending
Implementation	Next phase
Cloud and monitoring	Future phase

End of Unit 1 - AI-Powered Career & Skill Gap Analyzer